StutterClear Learning Guide

Resources to go deep on every part of the project and broad across the field: foundations, the papers behind each component, our own reports, and a 4-week reading order. All links are clickable.

A. Breadth: foundations

Topic	Resource	Why
Fourier intuition	3Blue1Brown: But what is the Fourier Transform?	20 min, the clearest visual intuition
DSP with code	Think DSP (free book)	Spectrograms, DCT, convolution, sampling, all in Python
DSP + DTW, rigorous	Mueller's FMP notebooks	Chapter 2 (Fourier) and chapter 3 (DTW), with derivations
Audio for ML	Valerio Velardo, The Sound of AI (YouTube)	MFCCs and mel spectrograms, explained for ML engineers
Speech ML, hands-on	Hugging Face Audio Course	Transformers for audio, ASR, TTS. **Start here.**
Speech processing, university level	Stanford CS224S	ASR, TTS and dialogue lectures
The textbook	Jurafsky & Martin, Speech and Language Processing (3rd ed. draft)	Phonetics, ASR and disfluency chapters
Deep learning engineering	Dive into Deep Learning	Optimisers, schedules, sequence models
Multi-GPU training	PyTorch DDP tutorial	What our teacher v2 uses on Kaggle
CTC / ASR alignment	Hannun, Sequence Modeling with CTC (Distill)	The dynamic-programming idea behind ASR
Real-time audio	Bencina, Real-time audio programming 101	Callbacks, buffers, why audio threads must never block
Reinforcement learning	Sutton & Barto (free book) and OpenAI Spinning Up	For the future cut/wait policy

B. Depth: the papers behind each part of StutterClear

Stuttering data and detection

- SEP-28k dataset: Lea et al. 2021
- Official SEP-28k-E speaker-exclusive split: Bayerl et al. 2022
- People who stutter and ASR, endpoint cut-offs: Lea et al. 2023 (Apple, CHI)
- **Current best detector, our main comparison:** Susac et al. 2026
- Synthetic frame labels + MIL (prior work to cite): Shih et al. 2024
- Frame-level detection: StutterCut, Ghosh et al. 2025
- On-device detector: Kozak 2026
- Forecasting stutters before they happen: Kumar, Daliri & Gupta 2026

Speech models

- wav2vec 2.0 -> HuBERT -> **WavLM (our teacher backbone)**
- SUPERB benchmark: how self-supervised layers are used for downstream tasks

Training tricks we use

- Knowledge distillation: Hinton et al. 2015

- Augmentation: SpecAugment
- Streaming decisions with a 'wait' output: Teaching BERT to Wait, Chen et al. 2022
- Weak-label event detection metrics: Mesaros et al. 2016 and the DCASE community

Labels and evaluation

- Verbatim ASR used for our silver labels and scoring: CrisperWhisper 2.0
- Verbatim vs intended transcripts on stuttered speech: Toyin et al. 2026
- Automatic naturalness score: UTMOS

Competing systems

- STEAMROLLER: ASR -> LLM -> cloned TTS, 3.8 s per segment
- SpeechAgent: its listening test is the risk we must beat

Streaming speech-to-speech and RL

- Moshi, Hibiki, Hibiki-Zero (GRPO to cut lag)
- GRPO itself: DeepSeekMath

Future phases (video, NVIDIA speech models)

- Audio + video stutter detection: Mohapatra et al. 2024
- Face landmarks: MediaPipe Face Landmarker
- NVIDIA speech models (Nemotron, Parakeet): NVIDIA NeMo on GitHub

Stuttering domain and ethics

- Stuttering Foundation
- Stuttering as a valid way of speaking: Tichenor, Constantino & Yaruss 2022
- Comfort and dignity over fluency: Cripping AI, FAccT 2026
- Delayed auditory feedback: Stuart et al. 1997

C. Our own documents (read these first)

- Real time stutter correction pipeline: the whole design space and the latency maths
- StutterClear novelty and SOTA check: what is new, what is patented, the best published numbers
- research_notes/: the detailed notes behind both reports, with a source for every claim

D. Reading order (about 4 weeks alongside building)

Week	Read	Goal
1	3Blue1Brown; HF Audio Course units 1-3; Think DSP spectrogram chapters; our two reports	Explain mel spectrograms, frames and hops, and the StutterClear pipeline end to end
2	wav2vec 2.0; WavLM; SEP-28k; Susac et al. 2026; Hinton distillation	Explain the teacher, MIL with log-sum-exp pooling, AUC vs F1, and why the student is causal
3	Bencina; Teaching BERT to Wait; CrisperWhisper 2.0; UTMOS; STEAMROLLER; SpeechAgent	Explain the latency budget, hysteresis cuts, crossfades, silver labels and how output is scored
4+	CS224S lectures as needed; Sutton & Barto and Mohapatra et al. when we reach RL and video	Be ready for any judge or reviewer question, and for the next phases